

Water Contamination Detection Report

Files

Baseline clean-water CSV: baseline_clean.csv

Input sensor CSV: input_sensor_data.csv

Result CSV: detection_results.csv

Input Columns

timestamp: time of the sensor reading

turbidity: cloudiness of water; sudden rise can indicate particles or microbes

orp: oxidation-reduction potential; sudden change can indicate chemical/biological activity

ph: acidity/alkalinity of water

conductivity: dissolved-ion level; sudden rise can indicate dissolved contaminants

temperature, label, notes: optional columns preserved if present

Clean-Water Calibration

turbidity: mean=0.447633, std_dev=0.039742

orp: mean=656.413300, std_dev=7.088657

ph: mean=7.258867, std_dev=0.043753

conductivity: mean=408.650583, std_dev=8.893090

VDA learned warning threshold: 2.821727

PAA warning threshold: area_ratio > 1.000000

Summary

Rows analyzed: 120

Rows with ALERT: 30

Rows with NO_ALERT: 90

VDA alerts: 30

PAA alerts: 30

How To Read One Warning

ALERT means at least one algorithm found the row unusual compared with clean water.

VDA warning: combined normalized distance is above the learned threshold.

PAA warning: radar/polygon area ratio is above 1.

Alert Rows (showing up to 50)

Row 41 at 2026-01-01T09:13:20: ALERT (VDA+PAA)

Values: turbidity=1.578, orp=622.065, ph=7.159, conductivity=478.353

Normalized: turbidity=28.442, orp=4.846, ph=2.282, conductivity=7.838

Decision: VDA distance=28.817 (threshold=2.822, alarm=True); PAA area ratio=97.423 (threshold=1.000, alarm=True)

Row 42 at 2026-01-01T09:13:25: ALERT (VDA+PAA)

Values: turbidity=1.599, orp=624.713, ph=7.211, conductivity=454.840

Normalized: turbidity=28.971, orp=4.472, ph=1.094, conductivity=5.194

Decision: VDA distance=28.715 (threshold=2.822, alarm=True); PAA area ratio=72.650 (threshold=1.000, alarm=True)

Row 43 at 2026-01-01T09:13:30: ALERT (VDA+PAA)

Values: turbidity=1.382, orp=622.298, ph=7.145, conductivity=484.964

Normalized: turbidity=23.511, orp=4.813, ph=2.602, conductivity=8.581

Decision: VDA distance=24.378 (threshold=2.822, alarm=True); PAA area ratio=87.438 (threshold=1.000, alarm=True)

Row 44 at 2026-01-01T09:13:35: ALERT (VDA+PAA)

Values: turbidity=1.472, orp=624.083, ph=7.149, conductivity=505.649

Normalized: turbidity=25.775, orp=4.561, ph=2.511, conductivity=10.907

Decision: VDA distance=27.229 (threshold=2.822, alarm=True); PAA area ratio=109.383 (threshold=1.000, alarm=True)

Row 45 at 2026-01-01T09:13:40: ALERT (VDA+PAA)

Values: turbidity=1.549, orp=612.846, ph=7.165, conductivity=506.333

Normalized: turbidity=27.713, orp=6.146, ph=2.145, conductivity=10.984

Decision: VDA distance=29.270 (threshold=2.822, alarm=True); PAA area ratio=127.868 (threshold=1.000, alarm=True)

Row 46 at 2026-01-01T09:13:45: ALERT (VDA+PAA)

Values: turbidity=1.359, orp=623.741, ph=7.209, conductivity=470.092

Normalized: turbidity=22.932, orp=4.609, ph=1.140, conductivity=6.909

Decision: VDA distance=23.244 (threshold=2.822, alarm=True); PAA area ratio=69.314 (threshold=1.000, alarm=True)

Row 47 at 2026-01-01T09:13:50: ALERT (VDA+PAA)

Values: turbidity=1.260, orp=616.875, ph=7.178, conductivity=485.816

Normalized: turbidity=20.441, orp=5.578, ph=1.848, conductivity=8.677

Decision: VDA distance=21.692 (threshold=2.822, alarm=True); PAA area ratio=79.431 (threshold=1.000, alarm=True)

Row 48 at 2026-01-01T09:13:55: ALERT (VDA+PAA)

Values: turbidity=1.494, orp=615.244, ph=7.203, conductivity=477.558

Normalized: turbidity=26.329, orp=5.808, ph=1.277, conductivity=7.748

Decision: VDA distance=26.900 (threshold=2.822, alarm=True); PAA area ratio=93.556 (threshold=1.000, alarm=True)

Row 49 at 2026-01-01T09:14:00: ALERT (VDA+PAA)

Values: turbidity=1.425, orp=615.552, ph=7.148, conductivity=479.606

Normalized: turbidity=24.592, orp=5.764, ph=2.534, conductivity=7.979

Decision: VDA distance=25.374 (threshold=2.822, alarm=True); PAA area ratio=93.200 (threshold=1.000, alarm=True)

Row 50 at 2026-01-01T09:14:05: ALERT (VDA+PAA)

Values: turbidity=1.273, orp=608.713, ph=7.160, conductivity=449.504

Normalized: turbidity=20.768, orp=6.729, ph=2.260, conductivity=4.594

Decision: VDA distance=21.195 (threshold=2.822, alarm=True); PAA area ratio=65.185 (threshold=1.000, alarm=True)

Row 51 at 2026-01-01T09:14:10: ALERT (VDA+PAA)

Values: turbidity=1.348, orp=594.912, ph=7.126, conductivity=477.958
Normalized: turbidity=22.655, orp=8.676, ph=3.037, conductivity=7.793
Decision: VDA distance=24.339 (threshold=2.822, alarm=True); PAA area ratio=105.782 (threshold=1.000, alarm=True)

Row 52 at 2026-01-01T09:14:15: ALERT (VDA+PAA)
Values: turbidity=1.468, orp=614.455, ph=7.148, conductivity=450.164
Normalized: turbidity=25.674, orp=5.919, ph=2.534, conductivity=4.668
Decision: VDA distance=25.717 (threshold=2.822, alarm=True); PAA area ratio=74.661 (threshold=1.000, alarm=True)

Row 53 at 2026-01-01T09:14:20: ALERT (VDA+PAA)
Values: turbidity=1.619, orp=620.160, ph=7.215, conductivity=457.647
Normalized: turbidity=29.474, orp=5.114, ph=1.003, conductivity=5.509
Decision: VDA distance=29.344 (threshold=2.822, alarm=True); PAA area ratio=80.944 (threshold=1.000, alarm=True)

Row 54 at 2026-01-01T09:14:25: ALERT (VDA+PAA)
Values: turbidity=1.378, orp=596.805, ph=7.199, conductivity=483.092
Normalized: turbidity=23.410, orp=8.409, ph=1.368, conductivity=8.371
Decision: VDA distance=25.008 (threshold=2.822, alarm=True); PAA area ratio=103.942 (threshold=1.000, alarm=True)

Row 55 at 2026-01-01T09:14:30: ALERT (VDA+PAA)
Values: turbidity=1.172, orp=614.479, ph=7.192, conductivity=445.331
Normalized: turbidity=18.227, orp=5.916, ph=1.528, conductivity=4.125
Decision: VDA distance=18.449 (threshold=2.822, alarm=True); PAA area ratio=49.586 (threshold=1.000, alarm=True)

Row 81 at 2026-01-01T09:16:40: ALERT (VDA+PAA)
Values: turbidity=0.696, orp=596.361, ph=7.048, conductivity=546.213
Normalized: turbidity=6.249, orp=8.472, ph=4.819, conductivity=15.468
Decision: VDA distance=17.892 (threshold=2.822, alarm=True); PAA area ratio=66.247 (threshold=1.000, alarm=True)

Row 82 at 2026-01-01T09:16:45: ALERT (VDA+PAA)
Values: turbidity=0.872, orp=600.843, ph=7.024, conductivity=585.672
Normalized: turbidity=10.678, orp=7.839, ph=5.368, conductivity=19.906
Decision: VDA distance=23.086 (threshold=2.822, alarm=True); PAA area ratio=111.298 (threshold=1.000, alarm=True)

Row 83 at 2026-01-01T09:16:50: ALERT (VDA+PAA)
Values: turbidity=0.750, orp=596.004, ph=7.018, conductivity=539.107
Normalized: turbidity=7.608, orp=8.522, ph=5.505, conductivity=14.669
Decision: VDA distance=17.907 (threshold=2.822, alarm=True); PAA area ratio=76.029 (threshold=1.000, alarm=True)

Row 84 at 2026-01-01T09:16:55: ALERT (VDA+PAA)
Values: turbidity=0.610, orp=610.013, ph=7.118, conductivity=515.694

Normalized: turbidity=4.085, orp=6.546, ph=3.220, conductivity=12.037
Decision: VDA distance=13.271 (threshold=2.822, alarm=True); PAA area ratio=33.936 (threshold=1.000, alarm=True)

Row 85 at 2026-01-01T09:17:00: ALERT (VDA+PAA)
Values: turbidity=0.630, orp=562.305, ph=7.109, conductivity=530.809
Normalized: turbidity=4.589, orp=13.276, ph=3.425, conductivity=13.736
Decision: VDA distance=18.564 (threshold=2.822, alarm=True); PAA area ratio=54.119 (threshold=1.000, alarm=True)

Row 86 at 2026-01-01T09:17:05: ALERT (VDA+PAA)
Values: turbidity=0.745, orp=580.621, ph=7.044, conductivity=518.236
Normalized: turbidity=7.482, orp=10.692, ph=4.911, conductivity=12.323
Decision: VDA distance=17.099 (threshold=2.822, alarm=True); PAA area ratio=71.306 (threshold=1.000, alarm=True)

Row 87 at 2026-01-01T09:17:10: ALERT (VDA+PAA)
Values: turbidity=0.752, orp=564.868, ph=7.046, conductivity=546.175
Normalized: turbidity=7.658, orp=12.914, ph=4.865, conductivity=15.464
Decision: VDA distance=20.631 (threshold=2.822, alarm=True); PAA area ratio=88.851 (threshold=1.000, alarm=True)

Row 88 at 2026-01-01T09:17:15: ALERT (VDA+PAA)
Values: turbidity=0.787, orp=581.756, ph=7.005, conductivity=543.191
Normalized: turbidity=8.539, orp=10.532, ph=5.802, conductivity=15.129
Decision: VDA distance=19.625 (threshold=2.822, alarm=True); PAA area ratio=92.002 (threshold=1.000, alarm=True)

Row 89 at 2026-01-01T09:17:20: ALERT (VDA+PAA)
Values: turbidity=0.711, orp=606.922, ph=7.088, conductivity=537.696
Normalized: turbidity=6.627, orp=6.982, ph=3.905, conductivity=14.511
Decision: VDA distance=16.428 (threshold=2.822, alarm=True); PAA area ratio=56.590 (threshold=1.000, alarm=True)

Row 90 at 2026-01-01T09:17:25: ALERT (VDA+PAA)
Values: turbidity=0.712, orp=575.162, ph=7.003, conductivity=532.941
Normalized: turbidity=6.652, orp=11.462, ph=5.848, conductivity=13.976
Decision: VDA distance=18.638 (threshold=2.822, alarm=True); PAA area ratio=79.494 (threshold=1.000, alarm=True)

Row 91 at 2026-01-01T09:17:30: ALERT (VDA+PAA)
Values: turbidity=0.774, orp=592.217, ph=7.078, conductivity=581.974
Normalized: turbidity=8.212, orp=9.056, ph=4.134, conductivity=19.490
Decision: VDA distance=21.991 (threshold=2.822, alarm=True); PAA area ratio=88.106 (threshold=1.000, alarm=True)

Row 92 at 2026-01-01T09:17:35: ALERT (VDA+PAA)
Values: turbidity=0.694, orp=585.181, ph=7.190, conductivity=502.659
Normalized: turbidity=6.199, orp=10.049, ph=1.574, conductivity=10.571

Decision: VDA distance=14.518 (threshold=2.822, alarm=True); PAA area ratio=40.070 (threshold=1.000, alarm=True)

Row 93 at 2026-01-01T09:17:40: ALERT (VDA+PAA)

Values: turbidity=0.708, orp=587.374, ph=7.058, conductivity=548.157

Normalized: turbidity=6.551, orp=9.739, ph=4.591, conductivity=15.687

Decision: VDA distance=18.688 (threshold=2.822, alarm=True); PAA area ratio=70.827 (threshold=1.000, alarm=True)

Row 94 at 2026-01-01T09:17:45: ALERT (VDA+PAA)

Values: turbidity=0.731, orp=590.126, ph=7.053, conductivity=555.427

Normalized: turbidity=7.130, orp=9.351, ph=4.705, conductivity=16.505

Decision: VDA distance=19.372 (threshold=2.822, alarm=True); PAA area ratio=76.502 (threshold=1.000, alarm=True)

Row 95 at 2026-01-01T09:17:50: ALERT (VDA+PAA)

Values: turbidity=0.599, orp=572.609, ph=7.050, conductivity=519.363

Normalized: turbidity=3.809, orp=11.822, ph=4.774, conductivity=12.449

Decision: VDA distance=16.809 (threshold=2.822, alarm=True); PAA area ratio=52.077 (threshold=1.000, alarm=True)